PROPERNESS IS EVENTUAL IN THE GOOD DIAGRAM SITUATION

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Editorial note. No omissions: the closing 'immediate from the construction' is the standard closed-subscheme-of-proper composition, written out as the displayed containment in the span.

Allowed standard prerequisites (editorial).

- Proper limit of proper fp. Limits, Lemma lemma-proper-limit-of-proper-finite-presentation (same chapter)
- Eventually proper. Limits, Lemma lemma-eventually-proper (same chapter)

Source reference. Original lemma 0CNR: sources/stacks-chow/tags.txt, source lines 15901–15901. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0CNS: sources/stacks-chow/tags.txt, source lines 15902–15902. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0CNP: sources/stacks-chow/tags.txt, source lines 15899–15899. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Lemma 0CNR. If f is proper, then there exists an $i_3 \geq i_0$ such that for $i \geq i_3$ we have f_i is proper.*

Proof. By the discussion in Remark 0CNS the choice of i_1 and W fitting into a diagram as in (0CNP) is immaterial for the truth of the lemma. Thus we choose W as follows. First we choose a closed immersion $X \to X'$ with $X' \to S$ proper and of finite presentation, see Lemma 09ZR. Then we choose an $i_3 \geq i_2$ and a proper morphism $W \to Y_{i_3}$ such that $X' = Y \times_{Y_{i_3}} W$. This is possible because $Y = \lim_{i \geq i_2} Y_i$ and Lemmas 01ZM and 081F. With this choice of W it is immediate from the construction that for $i \geq i_3$ the scheme X_i is a closed subscheme of $Y_i \times_{Y_{i_3}} W \subset S_i \times_{S_{i_3}} W$ and hence proper over Y_i. □

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